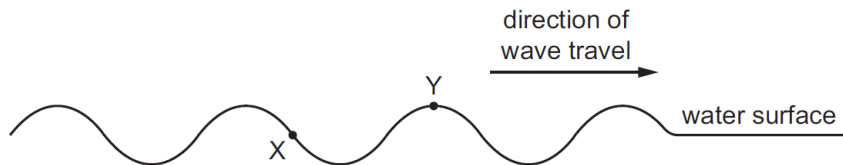


14. X and Y are two points on the surface of water in a ripple tank. A source of constant frequency generates a wave which travels past X and Y, causing them to oscillate vertically.



What is the phase difference between X and Y?

- A** 45° **B** 135° **C** 180° **D** 270°

Ans: D

From the diagram, path difference between X and Y, $\Delta x = 0.75\lambda$

Therefore, phase difference, $\Delta\phi_{XY} = \frac{\Delta x}{\lambda} \times 360^\circ = \frac{0.75\lambda}{\lambda} \times 360^\circ = 270^\circ$

15. A point source of sound emits energy equally in all directions at a constant rate and a